

Development of a Model for Real-Time Voice/Text to Sign Language & Gestures to Text Conversion

Problem Addressed

Communication between normal individuals and people with speech or hearing impairments is difficult because most people do not understand sign language. This creates barriers in education, workplaces, and daily social interactions. The paper proposes a system that enables communication by translating voice and text into sign language and converting gestures into text.

Key Contribution

The research develops a **bi-directional communication system** that performs:

- Voice → Text → Sign Language
- Text → Sign Language
- Hand Gesture → Text

This system improves accessibility and allows real-time communication between hearing and hearing-impaired individuals.

System Architecture

The proposed system consists of several components:

▪ Voice to Text Conversion

The **speech_recognition() library** is used to convert spoken words into text using Google speech API.

▪ Text to Sign Language

Each character of the input text is mapped to predefined sign language gesture images. These images are processed and displayed as **GIF animations representing the sign language sequence**.

▪ Real-Time Gesture Recognition

A webcam captures hand gestures, and **MediaPipe** is used to detect **21 hand landmarks**. These landmark coordinates are fed into an **Artificial Neural Network (ANN)** to classify the gesture and convert it into text.

Implementation

The system is implemented using several technologies:

- **Python** programming language
- **OpenCV** for image processing
- **MediaPipe** for hand landmark detection
- **TensorFlow / Keras ANN** for gesture classification
- **Tkinter GUI** for user interaction

The GUI allows users to input voice or text and view the corresponding sign language output.

Results

The system successfully demonstrates:

- Voice-to-sign language translation
- Text-to-sign language conversion
- Real-time gesture recognition

The results show that gestures such as “**Peace,**” “**Thumbs up,**” “**Stop,**” “**Rock,**” “**Smile,**” and **others** are accurately detected and translated into text using MediaPipe landmark detection and ANN classification

Conclusion

The proposed system provides an effective solution for bridging communication gaps between deaf and hearing individuals. By integrating **speech recognition, sign language generation, and gesture recognition**, the system enables real-time bidirectional communication and improves accessibility for people with disabilities.